

# 01 PROTOCOL

PRIVATE RECURRING CRYPTO SUBSCRIPTIONS FOR SOLANA

v0.9.9 · FROST RELEASE

Volta Team · Slashy Fx · Solo, full-time

Dogpatch · Superteam · April 2026

# Crypto rails leak. Card rails take 12%.

A merchant accepting subscriptions today picks one of two losing trades:

## Public chains

Solana Pay, Coinbase Commerce, every standard L1 rail.

**Every customer's wallet history is on display. The merchant sees it. The chain sees it. Anyone querying the chain sees it.**

## Card processors

Stripe, PayPal, Patreon, Apple / Google in-app.

**2.9% to 12% rake. The processor surveils every transaction. Chargebacks are a loaded gun the buyer holds.**

## No recurring privacy rail

Mixers (Tornado), shielded pools (Aztec, Zcash).

**Privacy primitives exist but no merchant SDK, no recurring billing, no Solana speed. Built for transfers, not subscriptions.**

# Three lines crossed in 18 months.

2024 Q3

## Compute budget bump

Solana raised the per-tx CU ceiling, making on-chain STARK verification economical (~900K CU, <0.001 SOL per verify). Custom FRI verifiers in 792 KB SBF binaries became feasible.

2024 AUG

## ML-KEM-768 finalised

NIST published the post-quantum KEM standard. Hybrid X25519 + ML-KEM-768 stealth addresses are now production-grade — privacy that survives a future quantum adversary.

2024-2026

## Creator-rail revolt

Patreon's 12% rake + Apple's 30% in-app + Stripe 2.9% — creators publicly leaving for crypto rails (Pump.fun, Hivemapper, indie SaaS). The market is asking for an exit.

The cryptography is ready. The chain is ready. The market is ready.

# Private recurring crypto subscriptions.

Drop-in SDK for any online merchant. Customer subscribes from the Protocol 01 wallet (or via QR scan). Merchant receives SOL or USDC at their own retailer wallet. Every link between the two is severed by ZK-STARK proofs and hybrid post-quantum stealth addresses.

**0.80%**

TOTAL ROUND-TRIP FEE

**0%**

MERCHANT PLATFORM CUT

**~10s**

ON-CHAIN SETTLEMENT

**PQ**

POST-QUANTUM HARDENED

## Three integration paths

P01-JS

Drop-in pay button + React widgets. 5 lines of code on the merchant's site.

MERCHANT-SDK

Server-side: register service, poll payments, watch vaults, issue access tokens.

AUTH-SDK

"Login with Protocol 01" — QR + biometric, gate by active subscription.

# Not a prototype. Real transactions, real signatures.

## 5 verified merchants live

Seeded on the on-chain registry program

QaQwpvBi1EQpevNE21D2oNBHFsLtoLwa7aXH26zRhQB

|                                                                                                    |             |
|----------------------------------------------------------------------------------------------------|-------------|
|  netflix-standard | 0.05 SOL/mo |
|  disney-plus      | 0.06 SOL/mo |
|  spotify-premium  | 0.03 SOL/mo |
|  youtube-premium  | 0.04 SOL/mo |
|  test-fast-sub  | 0.05 SOL/mo |

## 22/22 E2E checks pass on devnet

register → fetch → deregister, every field byte-equal, rent refunded on cleanup.

Net cost 0.00001 SOL.

```
// register sig
3odKFeJXyvpVcy7q6Df5tzqM627DE6gt5...

// deregister sig
4MPXSxxRyocawcGuYg93aWUVKj9WPvyx2...

// verify on Solana Explorer (devnet)
explorer.solana.com/?cluster=devnet
```

 tests passing across 17 SDKs and 14 Anchor programs ·  live-devnet write E2E ·  external-tarball install + import smoke

# Five lines on the merchant's server.

Frontend ships a drop-in widget. Backend runs a 30-second poll loop. That's it.

## Frontend — React

```
import { P01Provider, SubscriptionWidget }
  from '@protocol-01/p01-js/react';

<P01Provider config={{ merchantId: 'acme' }}>
  <SubscriptionWidget tiers={[
    { id: 'pro', price: 19.99, interval: 'monthly' }
  ]} />
</P01Provider>
```

## Backend — Node / Cloudflare

```
import { pollPaymentsForRetailer, issueAccessToken }
  from '@protocol-01/merchant-sdk';

setInterval(async () => {
  const r = await pollPaymentsForRetailer(c, retailer);
  r.forEach(p => grantAccess(p));
}, 30_000);
```

## What the merchant never sees

**HIDDEN** The customer's wallet identity

**HIDDEN** The customer's transaction history

**HIDDEN** Any link between two subscribers

# 14 programs. 1 custom STARK verifier.

## Our deployed programs

|          |                                       |
|----------|---------------------------------------|
| VERIFIER | p01_stark_verifier – FRI, 6 AIRs      |
| CORE     | zk_shielded – pool + sub vault        |
| CORE     | specter – stealth + streams           |
| CORE     | p01_registry – merchant directory     |
| CORE     | p01_zkspl – confidential SPL          |
| PQ       | p01_quantum_vault – WOTS+ + hash-lock |
| PQ       | p01_arcium – Cerberus MPC bridge      |
| RAILS    | p01_relayer – trustless relay         |
| RAILS    | p01_liquidity – instant unshield      |
| RAILS    | subscription / stream / fee-splitter  |
| RAILS    | p01_mugen – P2P fiat escrow           |

## Solana primitives leveraged

|                 |                                     |
|-----------------|-------------------------------------|
| Anchor 0.32     | DSL/framework, all 14 programs      |
| SPL Token       | USDC + native SOL handling          |
| PDAs            | deterministic registry / vault keys |
| Compute Budget  | 1.4M CU for STARK verify            |
| SystemProgram   | native SOL + account allocation     |
| Memo Program v2 | invoice memos for one-shots         |
| Helius RPC      | getProgramAccounts pagination       |
| Privy           | embedded wallet onboarding          |
| Arcium Cerberus | 9 MPC circuits, threshold compute   |

The custom FRI/STARK verifier fits in a **792 KB** SBF binary. ~900K CU per verification. No other L1 hosts a verifier of this scope at this cost.

# Built today, in 90 days, alone.

**14**

ANCHOR PROGRAMS

**17**

SDKS PUBLISHED

**6**

STARK AIRS

**2,400+**

TESTS PASSING

## Shipped

- 14 Anchor programs deployed on devnet, ~340 program tests
- Custom FRI/STARK verifier (792 KB SBF, 6 circuits)
- Mobile app v0.9.9 Frost (Android APK 119 MB)
- Chrome extension MV3 with on-device WASM prover
- Web marketing site · design doc · economic charter
- Mugen reference fiat <> crypto P2P (FROST escrow)
- 5 verified merchants on devnet registry

## Left before May 11

- External audit cycle (OtterSec / Neodyme — quote in flight)
- Mainnet deployment + first 5 paying merchants
- iOS TestFlight beta
- Atomic dust-to-stealth in `cancel_private_stark`

## Biggest risk

Audit timing. Privacy primitives without an audit are theatre. We do not deploy to mainnet without one full external cycle complete.



# 0.80% round-trip. Zero platform cut.

**0.30%**

SHIELD FEE

**0.50%**

UNSHIELD FEE

**0.80%**

ROUND-TRIP TOTAL

## Versus the rails merchants use today

| PROVIDER          | TAKE RATE | PRIVACY      | SETTLEMENT       |
|-------------------|-----------|--------------|------------------|
| PROTOCOL 01       | 0.80%     | NATIVE ZK    | ~10s on-chain    |
| Stripe (cards)    | 2.90%     | NONE         | 2-7 days         |
| Patreon           | 8-12%     | NONE         | Monthly          |
| Coinbase Commerce | 1.00%     | NONE         | ~30 min finality |
| BTCPay self-host  | 0.00%     | PSEUDONYMOUS | 10-60 min        |
| Solana Pay (raw)  | 0.00%     | NONE         | ~1s              |

**Worked example:** SaaS at €19.99/mo · 1,000 subs · 12 months · 10% churn · gross €143,449 → net to merchant **€142,668**. Same revenue on Stripe: ~€5,460 fee · on Patreon: €14,345 to €17,214 fee.

# One builder. Ninety days. Full-time.

## Slashy — sole builder, full-time

24 years old, based in Paris. Self-taught since 17. Epitech alumnus — Bachelor's Full-Stack Dev (2023), Master's Cybersecurity (2025) with specialisation offensive security + cryptography.

2 years at CaptNBoat (2023–2024) leading cybersecurity initiatives and product. Earlier shipped a production e-commerce platform end-to-end (2022). 18+ projects across web, mobile and backend before Protocol 01.

Full-time on Protocol 01 since January 2026. No team. No outside funding. No advisors with equity.

## Why this person

I read the FRI paper before writing the on-chain verifier. I chose the X25519 + ML-KEM-768 hybrid stealth construction myself — not copied from a reference. Every line of the cryptographic security boundary is mine: custom FRI verifier, Goldilocks-Poseidon AIRs, hybrid stealth, WOTS+ vault, MPC bridge.

The cybersecurity Master's means I ship without breaking the threat model. Seven years self-taught means I do not wait for anyone.

***"Financial privacy is a right, not a feature.  
No team, no funding, just execution."***

# What we need to ship.

The core is built. The cryptography is proven. The chain is ready. What stands between Protocol 01 and 100 paying merchants on mainnet:

## Audit funding

One full external cycle from OtterSec, Neodyme or Trail of Bits. Quote pending. ~\$80K-150K.

## First merchant cohort

5 to 10 Solana-native creators or SaaS to seed the verified registry on mainnet. Friend-of-Solana intros highest-leverage.

## Ecosystem partnerships

Helius (RPC), Privy (wallet onboarding), Arcium (MPC), Backpack (distribution). Conversations in flight.

**Privacy** is a right.  
**Recurring** is the rail.  
**Solana** is the chain.

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